

Dreaddit results, explained

What each number means, why it came out that way, and what the ACE playbooks learned

Qwen3-8B · v5 server panel · 100 test posts × 3 seeds · internal working results, not manuscript evidence

At a glance: three messages

Where the differences between methods come from on Dreaddit stress detection

+7–11 pp

ACE and GEPA clearly beat the base model

All four ACE conditions score 7.0–11.3 points above Base; GEPA, a label-driven prompt optimizer, +9.7. ICL, MIPROv2 and DC stay within 1.3 points of Base or fall below it.

46% → 59–86%

The gains come from recognising “no stress”

Base labels 73 of 100 posts as stressed; only 48 are. Most methods catch over 90% of stressed posts, but how often they correctly clear a non-stressed post (specificity) varies widely.

Acc ≠ grounding

Grounding (Conformability) moves separately from accuracy

MIPROv2 is the best grounded (97.7%) but only mid-pack on accuracy. Offline ACE with GT is near the top on accuracy yet the least grounded (76%), driven by one seed at 54%.

What was compared: task, model, methods

Every method uses the same model (Qwen3-8B) and the same 100 test posts. Only “what extra the model is given” changes.

Task

Read a Reddit post and decide whether the author expresses stress (0 = no, 1 = yes).

Data

100 test posts: 48 stressed, 52 not. A separate 100 for training and 100 for development.

Repetitions

Three runs per condition (seeds 42, 43, 44) on the same 100 posts; only order and optimizer randomness differ. The SD is run-to-run variation, not sampling error.

GT ✓ / ✗

Whether true labels are shown during adaptation. Scoring always uses the true labels.

Method	What the model is additionally given	When it learns
Base	Nothing beyond the task instruction	—
ICL	Eight labeled training examples in the prompt	offline
MIPROv2	Searched instruction + example combination with the best development F1	offline
GEPA	An instruction evolved by reflecting on failures, then selected	offline
ACE	A playbook: strategy bullets accumulated by generate → reflect → curate	offline / online
DC (CU)	A cheatsheet memory rewritten as a whole after every item	online

Terms

Offline = learn from the 100 training posts first, freeze, then test. Online = update after every test item (with GT ✓ the label is revealed only after the prediction is saved). ACE and DC build a memory; the others select a prompt.

ICL, MIPROv2 and GEPA need labels, so they run with GT ✓ only. The same Qwen3-8B plays every role (generation, reflection, curation).

How to read the metrics

What each number in the results table measures

Accuracy

Share of the 100 posts labeled correctly. A malformed (invalid) output counts as wrong.

How to read The most intuitive number, but it hides a one-sided labeling bias.

Macro-F1

The F1 score of each class (stressed / not stressed), averaged.

How to read Clearly below accuracy means one class is handled poorly.

Conformability (grounding)

A judge model scores whether the explanation and quotes are supported by the post on four dimensions (factual support, evidence relevance, attribution, no unsupported claims), 0/1/2 each. A pass needs 2 on all four.

How to read Independent of correctness. The judge is the same Qwen3-8B, so it is provisional.

$\Delta \text{Acc} \cdot \text{mean} \pm \text{SD}$

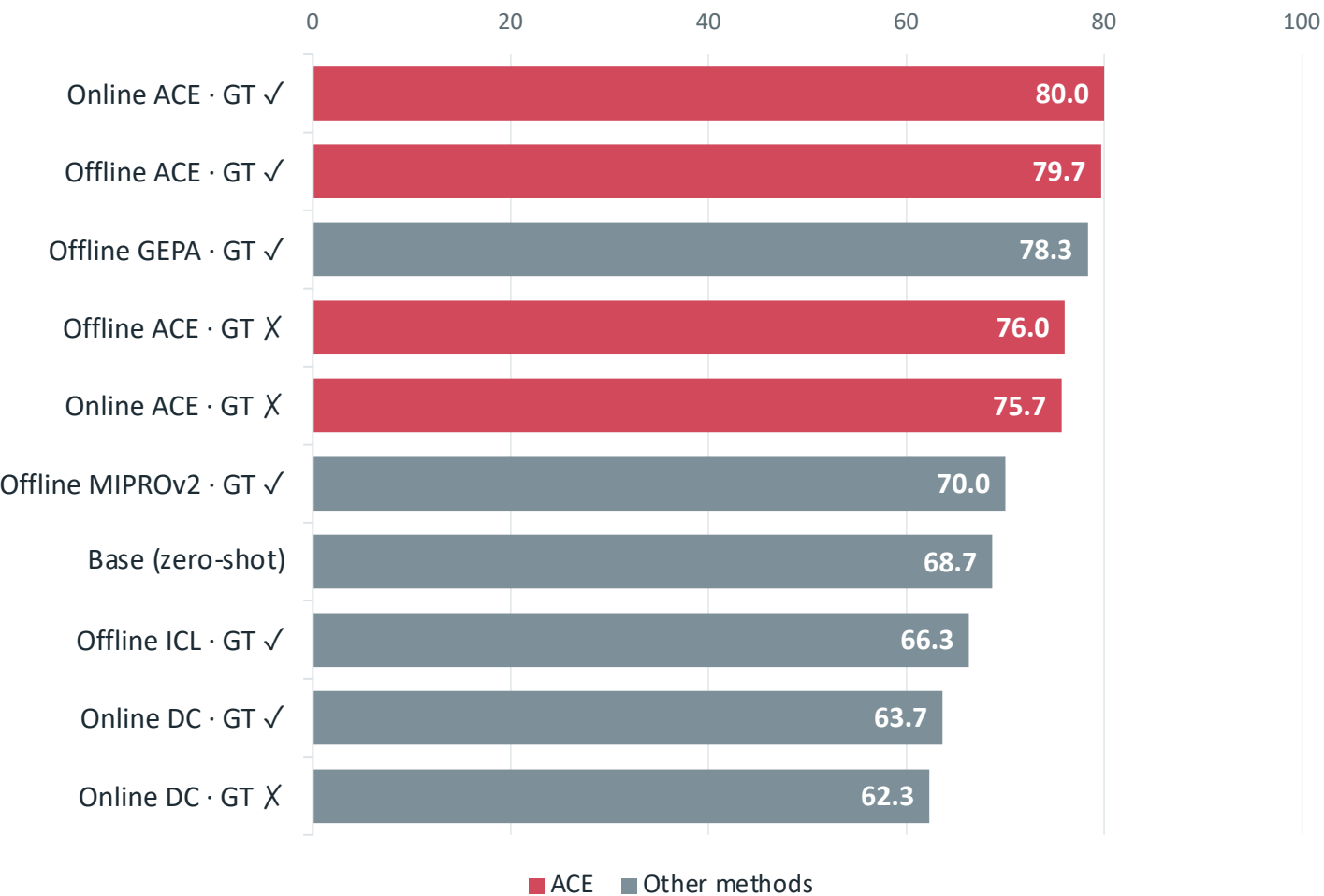
ΔAcc = difference from the Base mean, in points. $\pm$ is the SD over the three seeds.

How to read Same 100 posts solved three times, so the SD is how much a run wobbles.

Specificity (accuracy on non-stressed posts) and recall (accuracy on stressed posts) are examined on the error-analysis slide.

Result 1 · Accuracy: ACE and GEPA form the top group

Mean accuracy over three seeds on the 100 test posts (%)



What this means

All four ACE conditions reach 75.7–80.0%, 7–11 points above Base (68.7%).

GEPA reaches 78.3%, third overall and above both ACE conditions without labels.

The other prompt methods (ICL, MIPROv2) stay close to Base; ICL is 2.3 points lower.

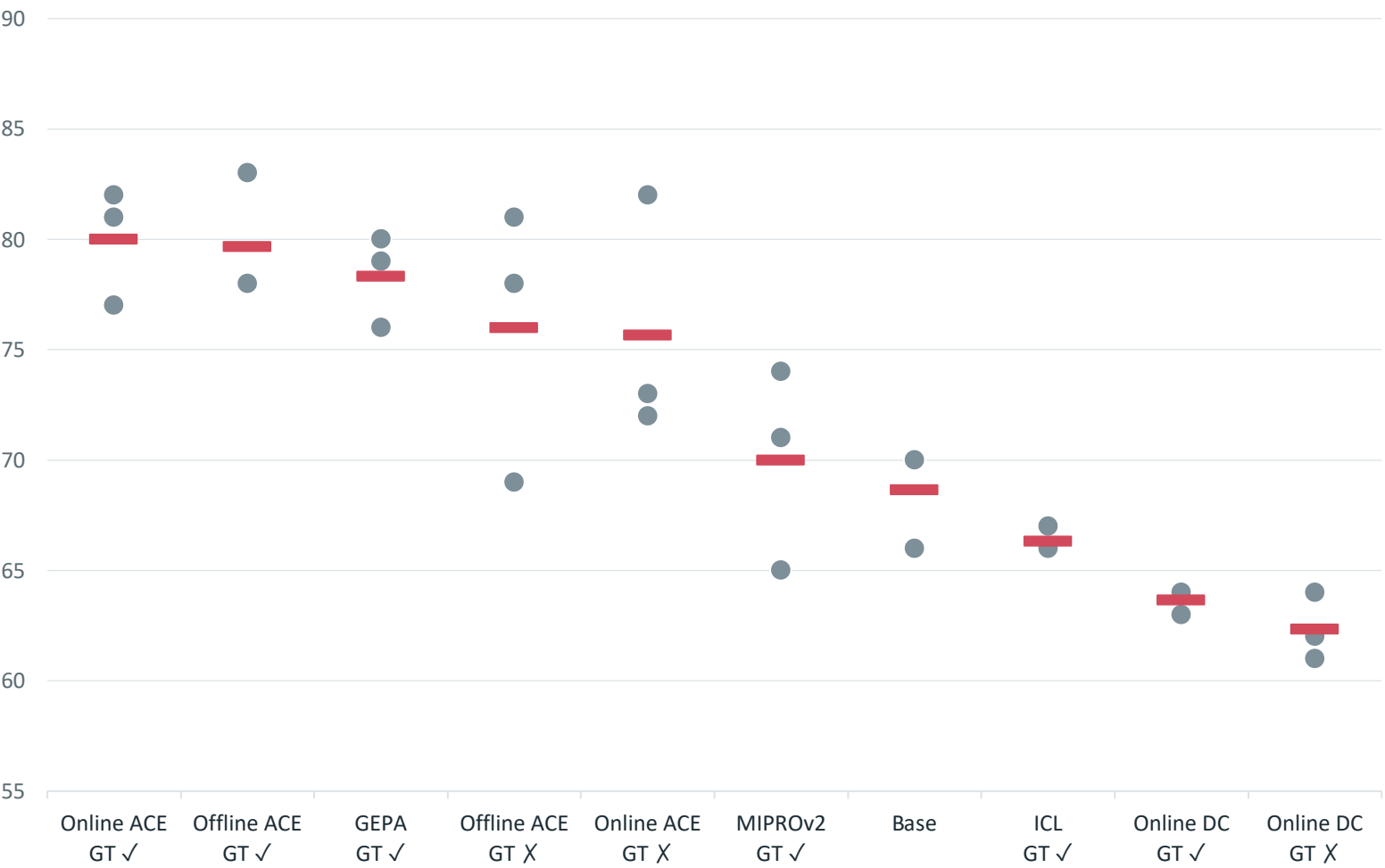
DC is 5–6 points below Base. Building a memory is not enough on its own; how it is built matters.

Next: how much these means move from seed to seed.

Invalid outputs count as wrong: three in Base seed 43, one in ICL seed 42.

Result 2 · Stability: one method can swing up to 12 points across seeds

Three dots = accuracy of seeds 42, 43, 44; red dash = mean



The vertical axis starts at 55%: this chart compares dot positions, not bar lengths.

What this means

Widest spread

Offline ACE · GT X: 69–81
Online ACE · GT X: 72–82
Offline MIPROv2 · GT ✓: 65–74

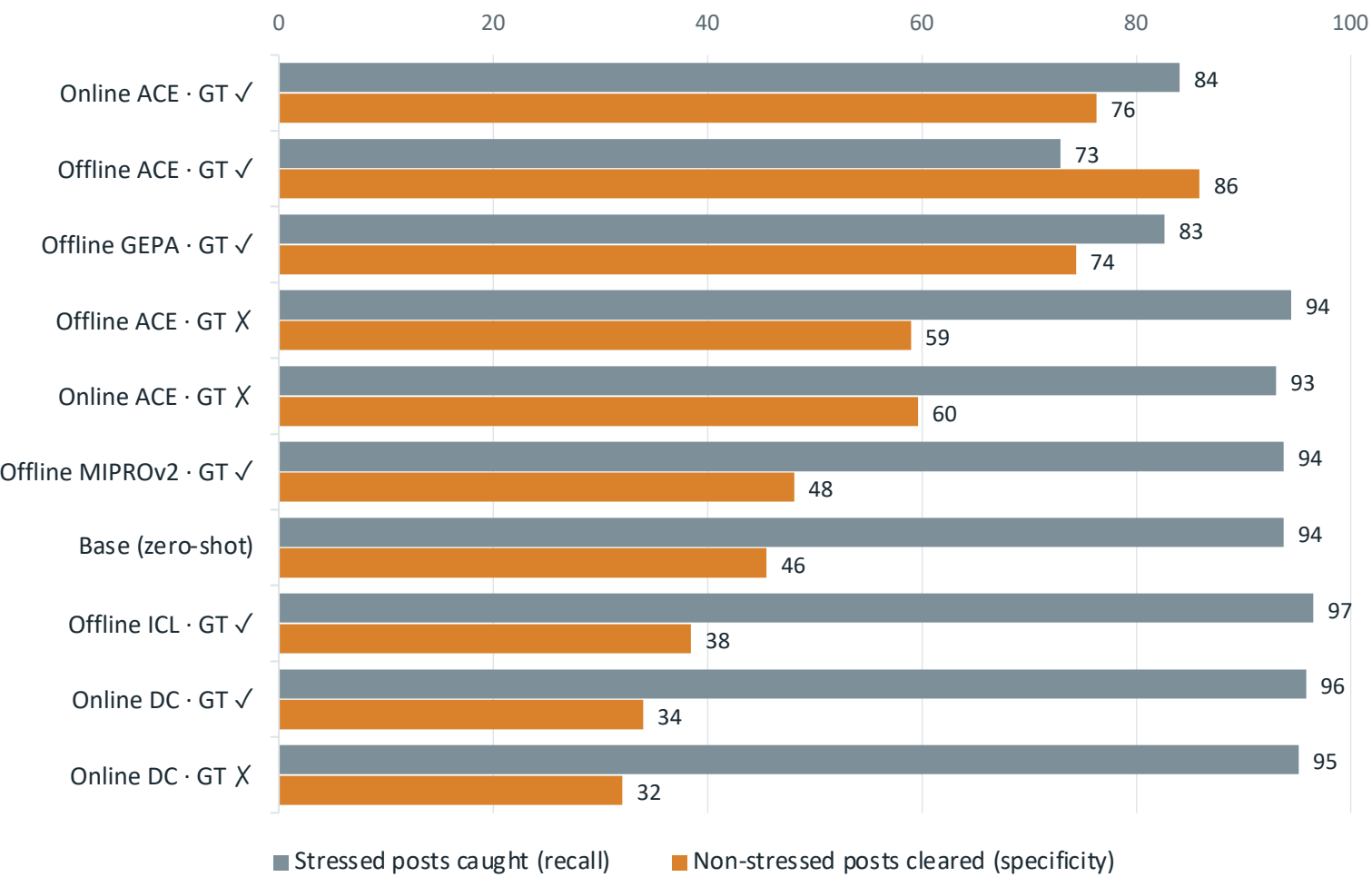
Narrowest spread

Online DC · GT X: 61–64
Offline ICL · GT ✓: 66–67
Online DC · GT ✓: 63–64

When the mean gain is smaller than the spread, as for MIPROv2 (+1.3), an effect is hard to claim. Even the worst ACE seed is at or above the Base mean.

Result 3 · Error analysis: the gap is about recognising “no stress”

Accuracy on stressed posts (recall, n = 48) and on non-stressed posts (specificity, n = 52), mean over three seeds



What this means

Base calls 73 of 100 posts stressed, though only 48 are.

So it catches nearly every stressed post (recall 94%) but misses more than half of the non-stressed ones (specificity 46%).

DC and ICL push this bias further (specificity 32–38%); ACE (59–86%) and GEPA (74%) reduce it.

Offline ACE with GT overshoots: specificity 86%, but recall falls to 73%.

Invalid outputs count as errors. Bar-end numbers are percentages.

Result 4 · A case study: the same method tipped the other way

Confusion matrices of Offline ACE with GT by seed. Only seed 43 swung towards “no stress”.

seed 42

	Pred: no	Pred: stress
True: no	33	19
True: stress	3	45

Accuracy 78% · recall 94% · specificity 63%
Grounding 90% (factual support 1.91/2, evidence relevance 1.89/2)

seed 43

	Pred: no	Pred: stress
True: no	52	0
True: stress	22	26

Accuracy 78% · recall 54% · specificity 100%
Grounding 54% (factual support 1.54/2, evidence relevance 1.52/2)

seed 44

	Pred: no	Pred: stress
True: no	49	3
True: stress	14	34

Accuracy 83% · recall 71% · specificity 94%
Grounding 84% (factual support 1.87/2, evidence relevance 1.83/2)

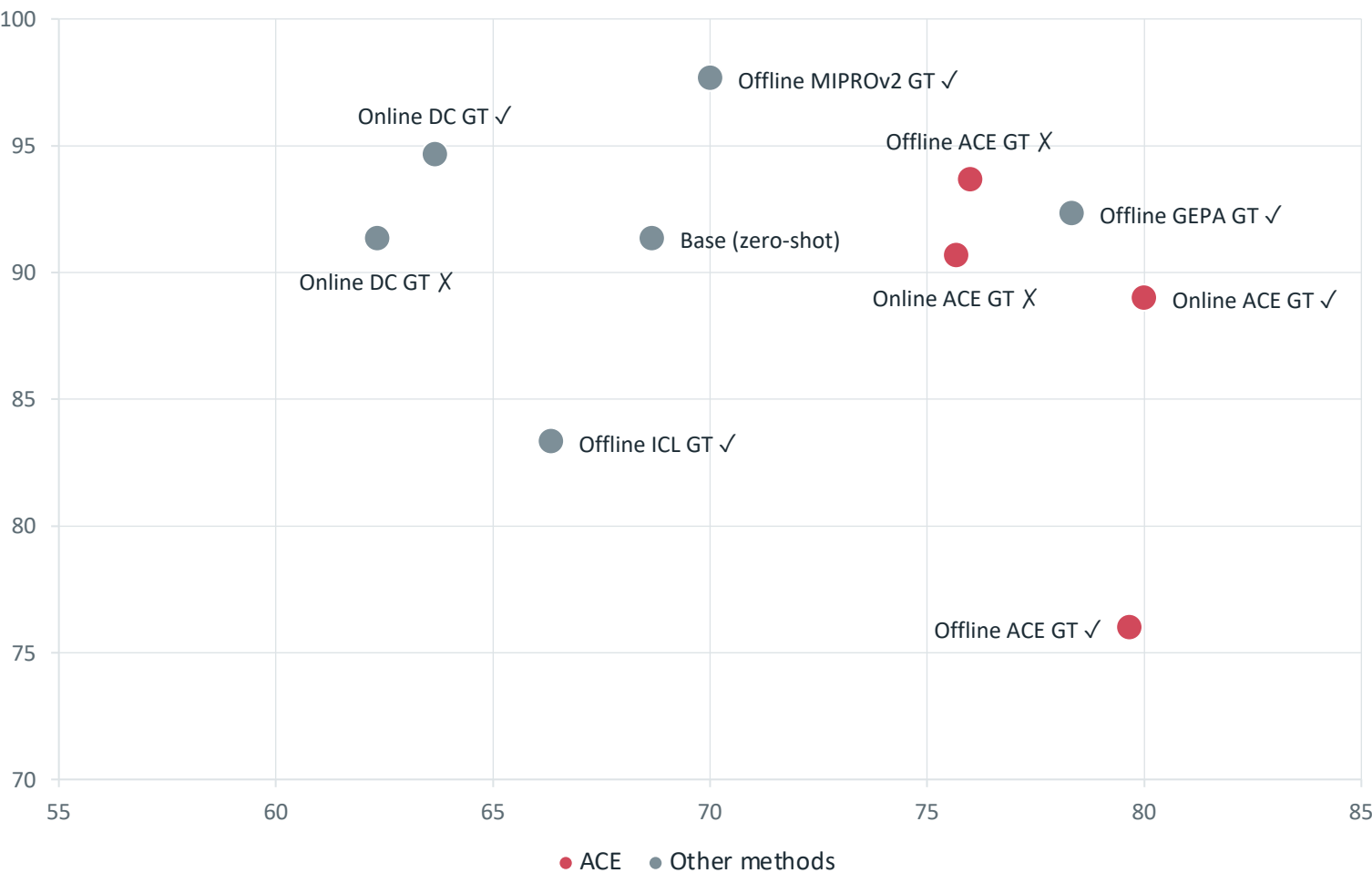
What this means

The seed-43 playbook learned to avoid calling stress: false positives dropped to zero, but 22 stressed posts were missed. Accuracy still looks fine at 78%, yet grounding collapsed to 54% (factual support and evidence relevance fell). When labeling tips to one side, the quality of explanations and quotes falls with it; the three-seed mean (76 ± 19) hides this.

Cell values are numbers of posts. True “no” = 52 posts, true “stress” = 48 posts.

Result 5 · Grounding and accuracy move independently

Horizontal = accuracy, vertical = Conformability (grounding), means over three seeds



What this means

Up and to the right is better, but the two axes do not rise together.

MIPROv2: best grounding (97.7%), middle accuracy (70.0%).
Offline ACE with GT: high accuracy (79.7%), lowest grounding (76%, driven by seed 43).
GEPA: high on both (78.3%, 92.3%).

Grounding differences come only from factual support and evidence relevance; attribution and no-unsupported-claims are at the maximum in nearly every run.

→ Report both accuracy and grounding.

Conformability is scored by the same Qwen3-8B (provisional, no human calibration). The judge sees neither the method nor the true label.

How to read an ACE playbook: what accumulated (seed 42)

A playbook is a list of strategy bullets the model writes and edits itself; each bullet carries helpful/harmful votes

OFFLINE • GT ✓

36 bullets

65 helpful votes · 1 flagged harmful

OFFLINE • GT ✗

63 bullets

107 helpful votes · 2 flagged harmful

ONLINE • GT ✓

51 bullets

68 helpful votes · 5 flagged harmful

ONLINE • GT ✗

50 bullets

84 helpful votes · 0 flagged harmful

Anatomy of one bullet

```
[calc-00001] helpful=26 harmful=0 :: Stress Level = ...
```

ID — section code (calc = FORMULAS & CALCULATIONS) plus a number. IDs never change.

helpful / harmful — +1 each time the Reflector judges the bullet helpful or harmful on an item.

Content — a strategy added by the Curator. Existing bullets are never deleted (ADD only).

What this means

198 of 200 bullets sit in one section, FORMULAS & CALCULATIONS.

ACE's default template comes from finance tasks, so its sections are named for formulas and calculations; Qwen wrote even its stress heuristics as "formulas".

The same five-factor stress formula appears nine times (Offline GT ✓).

The "refine" half of ACE's grow-and-refine (deduplication) is off in this configuration, so near-duplicates keep piling up.

Offline playbooks are built from the 100 training posts and then frozen; online playbooks are the final state after updating through the 100 test posts.

Offline playbooks: with labels it learns rules, without labels it learns cues

Representative bullets (top helpful) from playbooks built on the 100 training posts, then frozen

GT ✓ · learns from labels (accuracy 79.7%)

[calc-00001] helpful 26 · harmful 0

Stress Level Calculation: Stress Level = (Uncertainty Score + Seeking Advice Score + Decision Complexity Score) / 3. If the result is ≤ 1.5 , classify as 0 (no stress); if ≥ 2.5 , classify as 1 (stress).

[ctx-00002] helpful 8 · harmful 10 (flagged)

Identify stress through emotional language and contextual cues, such as phrases like “...” or “...”, which indicate emotional distress and anxiety.

GT ✗ · self-reflection only (accuracy 76.0%)

[calc-00001] helpful 37 · harmful 0

Identify indirect signs of stress such as uncertainty, indecision, and seeking advice. These can be strong indicators of stress even if the author does not explicitly state it.

[calc-00016] helpful 20 · harmful 0

When the author explicitly expresses self-blame and self-hatred, prioritize these as the strongest indicators of stress, as they directly reflect emotional distress and align with the playbook's guidance to focus on the most ...

What this means

The GT ✓ playbook writes decision rules such as “add the scores; at or below 1.5 means no stress”, although the task never asked for a score. The GT ✗ playbook accumulates qualitative guidance such as “watch for indirect signs (uncertainty, seeking advice)”. Both beat Base, but the rule-based playbook can overshoot to one side, as seed 43 did.

Seed 42 for every condition, fixed before viewing. Bullet text verbatim; multi-word post quotations replaced with “...”; long bullets truncated (...).

Online playbooks: with labels, a “no stress” rule appears

Representative bullets from the final playbooks after updating through the 100 test posts

GT ✓ · label revealed after each prediction specificity 76% · recall 84%

[calc-00015] helpful 22 · harmful 3 (flagged)

Use a binary classification system where the presence of any explicit stress indicators, including events that 'blindsided' the author or indicate lack of preparedness, is sufficient to classify the text as 'stress'. This approach prioritizes the ...

[calc-00019] helpful 8 · harmful 11 (flagged)

Use a binary classification system where the presence of any explicit stress indicators such as frustration, fatigue, anger, or sensory overload is sufficient to classify the text as 'stress'. The absence of these indicators, even if other emotional ...

[calc-00032] helpful 6 · harmful 0

Use a simplified binary classification system where the absence of explicit stress indicators (e.g., frustration, anger, sensory overload) is sufficient to classify the text as 'no stress'. This approach avoids overcomplication and ensures alignment ...

GT ✗ · no labels specificity 60% · recall 93%

[calc-00019] helpful 25 · harmful 0

Ensure that the final label always matches the reasoning and that the evidence supports the label. If multiple stress indicators are present, classify as 'stress' (label 1) regardless of the author's denial or minimization.

[ctx-00001] helpful 15 · harmful 0

Always reference playbook bullets when available to ensure that the reasoning is aligned with the expected framework and to improve the accuracy of the answer.

[calc-00014] helpful 9 · harmful 0

Prioritize the presence of multiple stress indicators over any denial or minimization by the author. If multiple stress indicators are present, classify as 'stress' (label 1) regardless of the author's denial.

What this means

Alongside its stress rules, the GT ✓ playbook adds “no explicit indicators means no stress” (calc-00032), and its broad stress rule (calc-00019) collected more harmful than helpful votes (11 vs 8): label feedback penalised over-calling. The GT ✗ playbook leans on “call stress even if the author denies it”. Specificity points the same way: 76% vs 60% (a correlation; causality is not yet tested).

Pink cards = bullets with at least one harmful vote. Seed 42 for every condition, fixed before viewing. Bullet text verbatim; multi-word post quotations replaced with “...”; long bullets truncated (...).

Takeaways and questions for discussion

Takeaways

- 1 ACE's gain comes mostly from learning to label non-stressed posts as non-stressed (specificity 46% → 59–86%).
- 2 Rule-like playbooks (GT ✓) are powerful but can overshoot (seed 43: zero false positives, 22 missed stressed posts).
- 3 Grounding is separate from accuracy, and it collapsed together with the skewed labeling in that run.
- 4 Playbook structure is not what the paper envisions: one section absorbs everything and one formula repeats nine times.
- 5 A strong prompt optimizer comes close: GEPA gains +9.7 with grounding 92%, so ACE's edge over GEPA is at most 1.7 points here.

Questions

- 1 Which bullets in the seed-43 playbook pushed towards “no stress”?
- 2 Does enabling the refine (deduplication) step reduce the skew and repetition?
- 3 Does the grounding ranking hold with a judge from a different model family?
- 4 AppWorld shows the same one-section pileup. Is that ACE's general failure mode on small models?